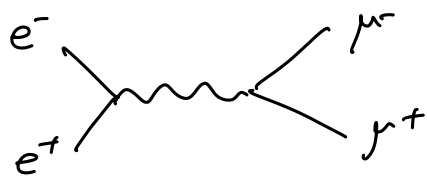
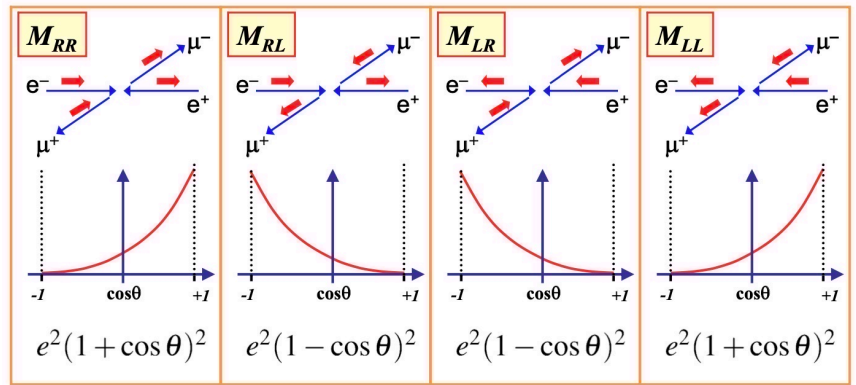


$e^+e^- \rightarrow \mu^+\mu^-$ cross section



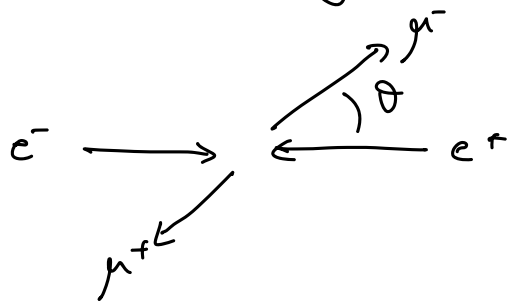
$$\frac{d\sigma}{d\Omega} = \frac{1}{64\pi^2} \frac{1}{s} |M|^2$$



4 initial states $S_z = \pm 1$, $S_x = 0$.

4 final states $S_z = \pm 1$, $S_x = 0$.

only contributions



$M_{RR}: e_R^- \bar{\nu}_R$

$M_{RL}: e_R^- \bar{\nu}_L$

$M_{LR}: e_L^- \bar{\nu}_R$

$M_{LL}: e_L^- \bar{\nu}_L$

average over initial states and sum over final states

$$\langle |M|^2 \rangle = \frac{1}{4} (|M_{RR}|^2 + |M_{RL}|^2 + |M_{LL}|^2 + |M_{LR}|^2)$$

$$= \frac{1}{4} [2(1+\cos\theta)^2 + 2(1-\cos\theta)^2] (4\pi\alpha)^2$$

$$\alpha = \frac{e^2}{4\pi}$$

$$= \frac{1}{4} [4 + 4\cos^2\theta] (4\pi\alpha)^2$$

$$\frac{d\sigma}{d\Omega} = \frac{1}{64\pi^2} \frac{1}{s} \frac{1}{4} 4(1+\cos^2\theta) (4\pi\alpha)^2$$

$$= \frac{16\alpha^2}{64\pi} \frac{1}{s} (1+\cos^2\theta)$$

$$\sigma = \int \left(\frac{d\sigma}{d\Omega} \right) d\Omega$$

$$d\Omega = d\phi \sin\theta d\theta$$

$$\sigma = \int_0^{2\pi} d\varphi \int_0^\pi \frac{d\sigma}{d\Omega} \sin\theta d\theta = 2\pi \frac{\alpha^2}{4} \frac{1}{s} \int_0^\pi (1 + \cos^2\theta) \sin\theta d\theta$$

$$\int_{-1}^1 (1+x^2) dx$$

$$\sigma = \frac{16\pi}{3} \frac{1}{s} \alpha^2$$

$S = 4E^2$ E : energy of the beam.

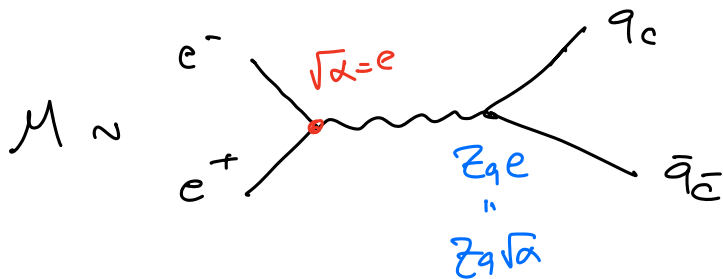
$\hbar c = 200 \text{ MeV} \cdot \text{fm}$.

E : in GeV $\alpha \approx \frac{1}{137}$

$$\sigma = \frac{86.8 \text{ nb}}{S [\text{GeV}]^2} = \frac{86.8 \text{ nb}}{4 E_{\text{beam}}^2 [\text{GeV}]^2} \quad e^+e^- \rightarrow \mu^+\mu^-$$

how to compute $e^+e^- \rightarrow \text{hadrons}$? in QED

$e^+e^- \rightarrow q_c \bar{q}_c$ c : color q : flavor



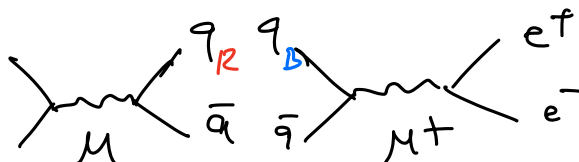
$$\sigma(e^+e^- \rightarrow q_c \bar{q}_c) \propto \left(\begin{smallmatrix} ? \\ 0 \end{smallmatrix} \right) \sigma(e^+e^- \rightarrow \mu^+\mu^-)$$

Fix flavor $\Rightarrow$ still have N_c colors

$$M = \begin{array}{c} q_c \\ \diagup \quad \diagdown \\ \text{---} \text{---} \text{---} \\ \diagdown \quad \diagup \\ q_c \end{array} = \begin{array}{c} q_R \\ \diagup \quad \diagdown \\ \text{---} \text{---} \text{---} \\ \diagdown \quad \diagup \\ \bar{q}_R \end{array} + \begin{array}{c} q_B \\ \diagup \quad \diagdown \\ \text{---} \text{---} \text{---} \\ \diagdown \quad \diagup \\ \bar{q}_B \end{array} + \begin{array}{c} q_G \\ \diagup \quad \diagdown \\ \text{---} \text{---} \text{---} \\ \diagdown \quad \diagup \\ \bar{q}_G \end{array}$$

$$|M_{q\bar{q}}|^2 = |M_{R\bar{R}}|^2 + |M_{G\bar{G}}|^2 + |M_{B\bar{B}}|^2 \quad (3 \text{ colors})$$

$$|M|^2 = M^\dagger M$$



$$\sigma(e^+e^- \rightarrow q_c \bar{q}_c) = N_c \cdot Z_q^2 \sigma(e^+e^- \rightarrow \mu^+ \mu^-)$$

$$\sigma(e^+e^- \rightarrow \text{hadrons}) = \sum_{\text{flavors}} N_c Z_q^2 \sigma(e^+e^- \rightarrow \mu^+ \mu^-)$$

$$R = \frac{\sigma(e^+e^- \rightarrow \text{hadrons})}{\sigma(e^+e^- \rightarrow \mu^+ \mu^-)} = N_c \sum_i^{\text{flav}} Z_{q_i}^2$$

$$\text{flavors: } \sqrt{s} \geq 2m_{q_i}$$

$$\text{measure experimentally } R = \frac{N(e^+e^- \rightarrow \text{had})}{N(e^+e^- \rightarrow \mu^+ \mu^-)}$$

$$\textcircled{1} \quad \sqrt{s} \geq 2m_u, 2m_d, 2m_s \quad \sim \quad \sqrt{s} \approx \text{few GeV}$$

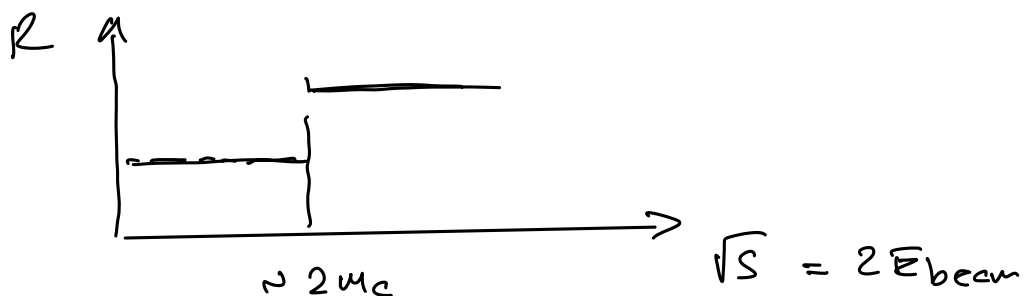
$$R = N_c \cdot \left[\underbrace{\left(\frac{2}{3}\right)^2}_{Z_u} + \underbrace{\left(-\frac{1}{3}\right)^2}_{Z_d} + \underbrace{\left(-\frac{1}{3}\right)^2}_{Z_s} \right] = \frac{4+1+1}{9} N_c$$

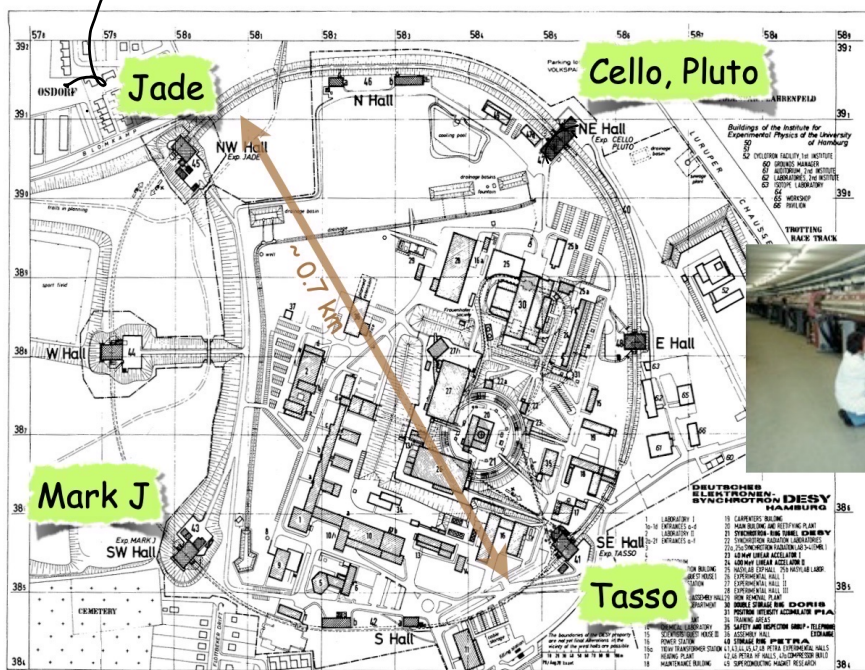
$$= \frac{6}{9} N_c = \frac{6}{3} = 2 \text{ if 3 colors.}$$

$$\sqrt{s} \geq 2m_c$$

$$R = 3 \cdot \left[\left(\frac{2}{3}\right)^2 + \left(-\frac{1}{3}\right)^2 + \left(-\frac{1}{3}\right)^2 + \left(\frac{2}{3}\right)^2 \right] =$$

$$= 3 \cdot \frac{4+1+1+4}{9} = 3 \cdot \frac{10}{9} = \frac{10}{3}$$





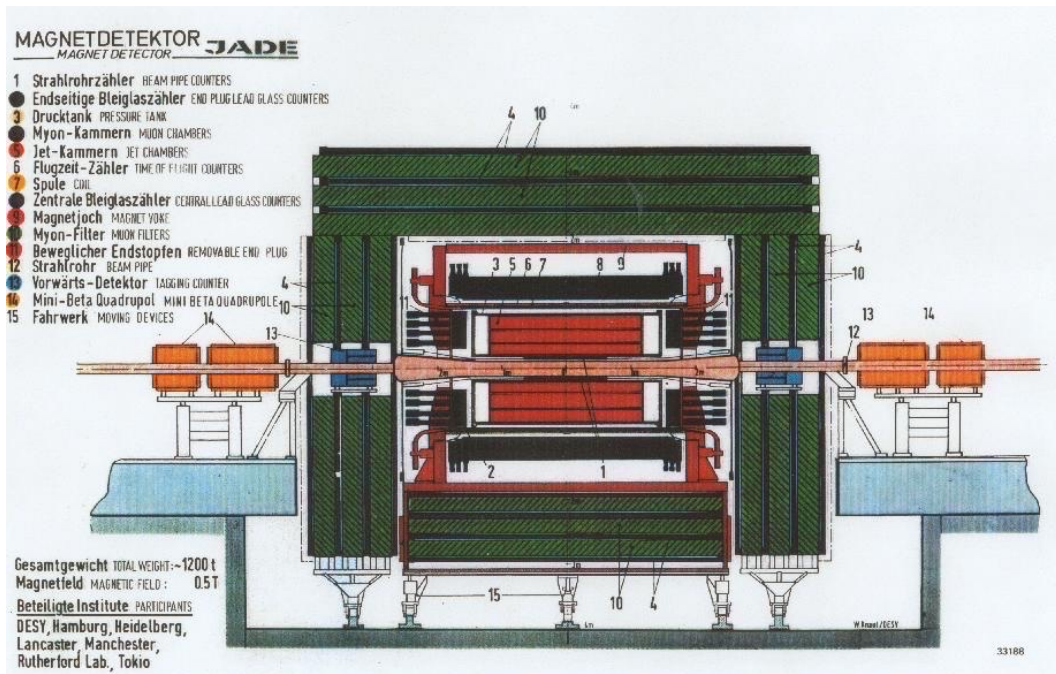
PETRA @ DESY

Hamburg

$\sqrt{s}$

$12 \rightarrow 46.7 \text{ GeV}$

1976 $\rightarrow$ 1986



Japan

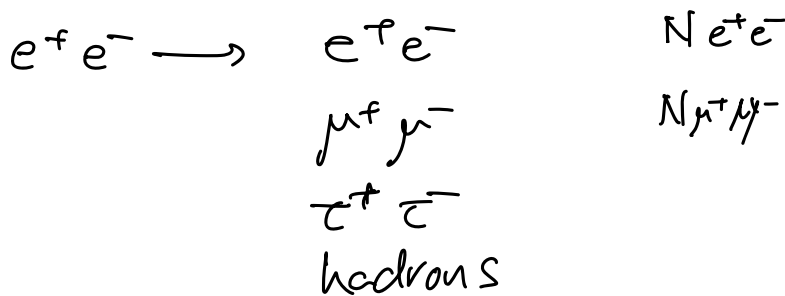
Deutschland

England

Detector

$B = 0.5 \text{ Tesla}$

$P = 0.3 \text{ B(T) R(M)}$



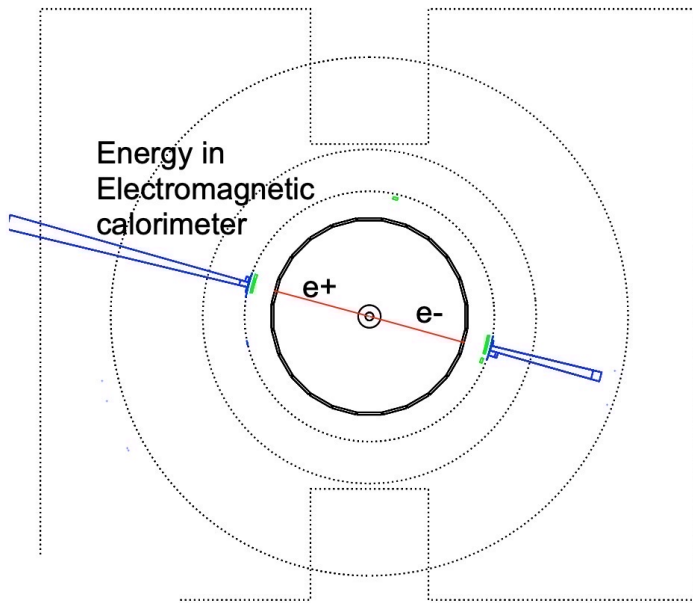
$m_\tau = 1.78 \text{ GeV}$

$$N_{e^+e^-} = \sigma(e^+e^- \rightarrow e^+e^-) \cdot L_{\text{int}} \cdot \Delta t$$

$$N_{\mu\mu} = \sigma(e^+e^- \rightarrow \mu^+\mu^-) \cdot L_{\text{int}} \cdot \Delta t$$

$$N_{\text{had}} = \sigma(e^+e^- \rightarrow \text{had}) \cdot L_{\text{int}} \cdot \Delta t$$

$$e^+ e^-$$

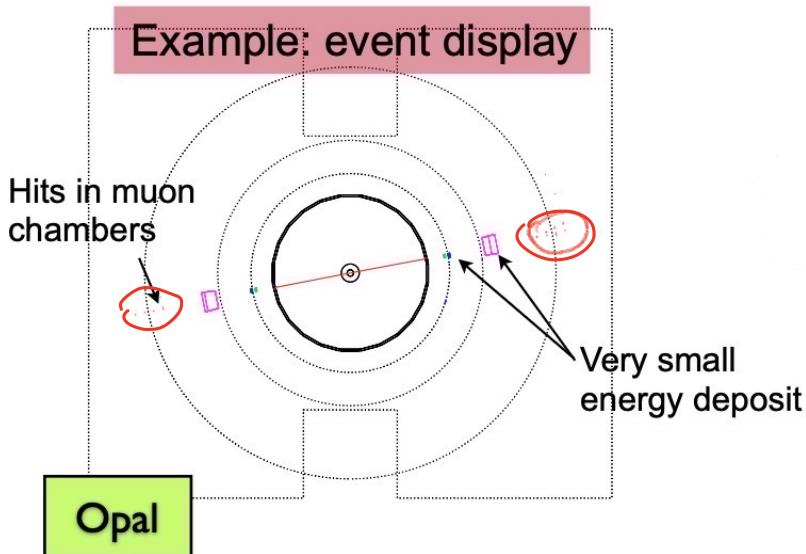


2 charged tracks
in trackers

large EM deposits
in EM Calorimeter

$$\mu^+ \mu^-$$

Example: event display



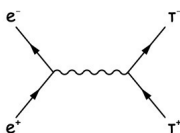
2 charged tracks

2 very small ener.
depos. in EM Cal.

small dep. also
in had. Cal.

Signal in Muon Chambers.

$$\tau^+ \tau^-$$

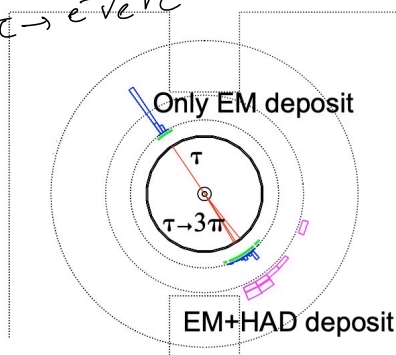


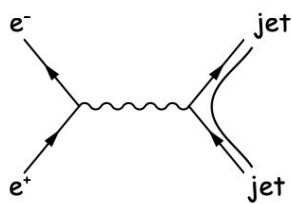
$$\tau^- \rightarrow e^- \bar{\nu}_e \nu_\tau$$

$$\mu^- \bar{\nu}_\mu \nu_\tau$$

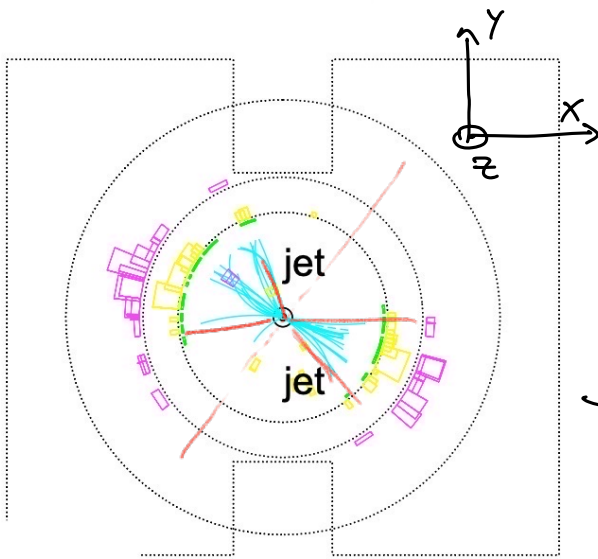
$$3\pi$$

$$\tau \rightarrow e^- \bar{\nu}_e \nu_\tau$$





hadronic final state
 many tracks.
 many EM deposits
 many had calo deposits



jets of hadrons

measure R from N_{ee} , $N_{\mu\mu}$, N_{had} .

$$R = \frac{\sigma(e^+e^- \rightarrow had)}{\sigma(e^+e^- \rightarrow \mu^+\mu^-)}$$

$$N_{ee} = \sigma_{ee} \cdot L_{inst} \cdot \Delta t$$

$$N_{\mu\mu} = \sigma_{\mu\mu} \cdot L_{inst} \cdot \Delta t$$

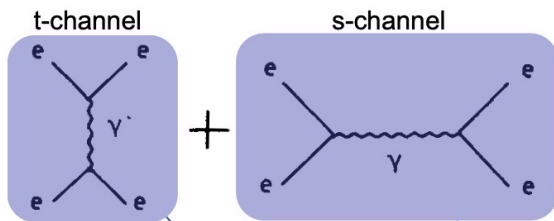
$$N_{had} = \sigma_{had} \cdot L_{inst} \cdot \Delta t$$

$$L_{inst} \cdot \Delta t = \frac{N_{ee}}{\sigma_{ee}(theory)}$$

$$\sigma_{had} = \frac{N_{had}}{L_{inst} \Delta t} = \frac{N_{had}}{N_{ee}} \sigma_{ee}(theory)$$

$$\sigma_{\mu\mu} = \frac{N_{\mu\mu}}{N_{ee}} \sigma_{ee}(theory) \Rightarrow \text{measure } R.$$

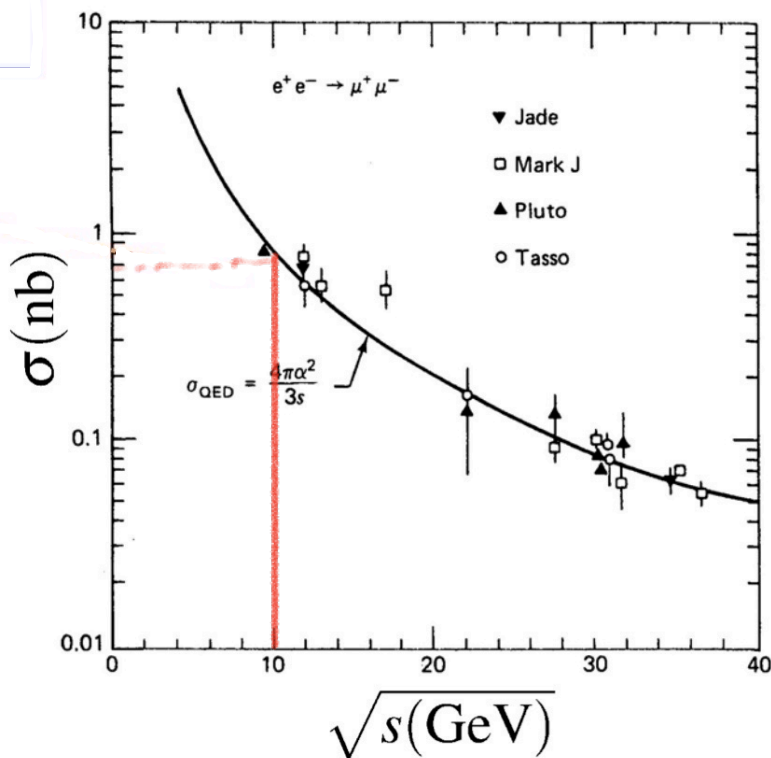
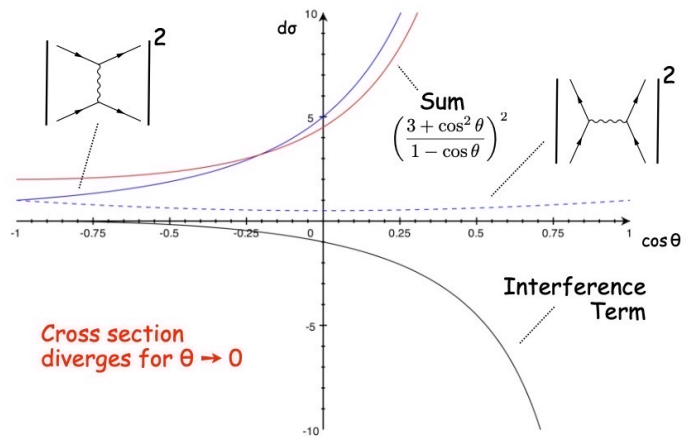
Bhabha scattering
 well calculated
 in QED



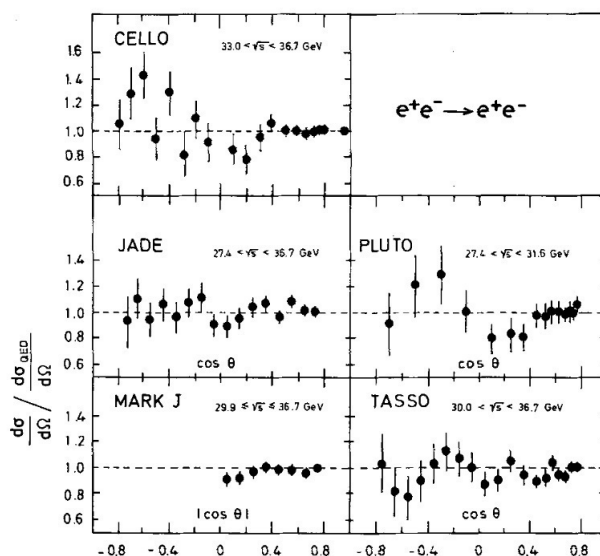
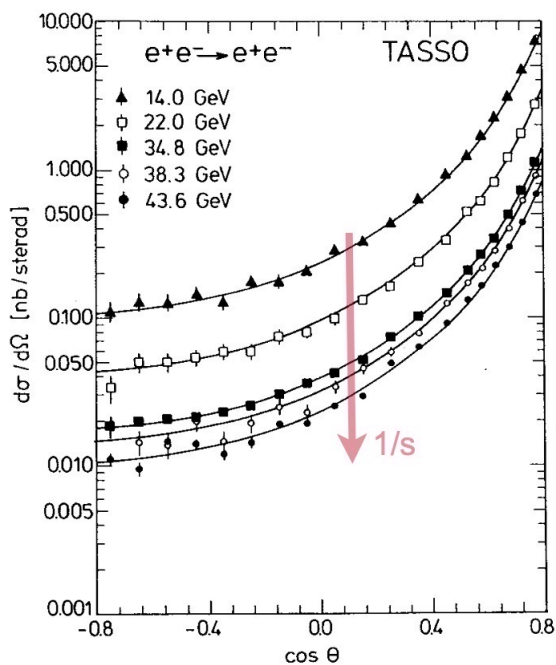
$$\frac{d\sigma_0}{d\Omega} = \frac{\alpha^2}{2s} \left(\frac{q'^4 + s^2}{q^4} + \frac{2q'^4}{q^2 s} + \frac{q'^4 + q^4}{s^2} \right)$$

interference term

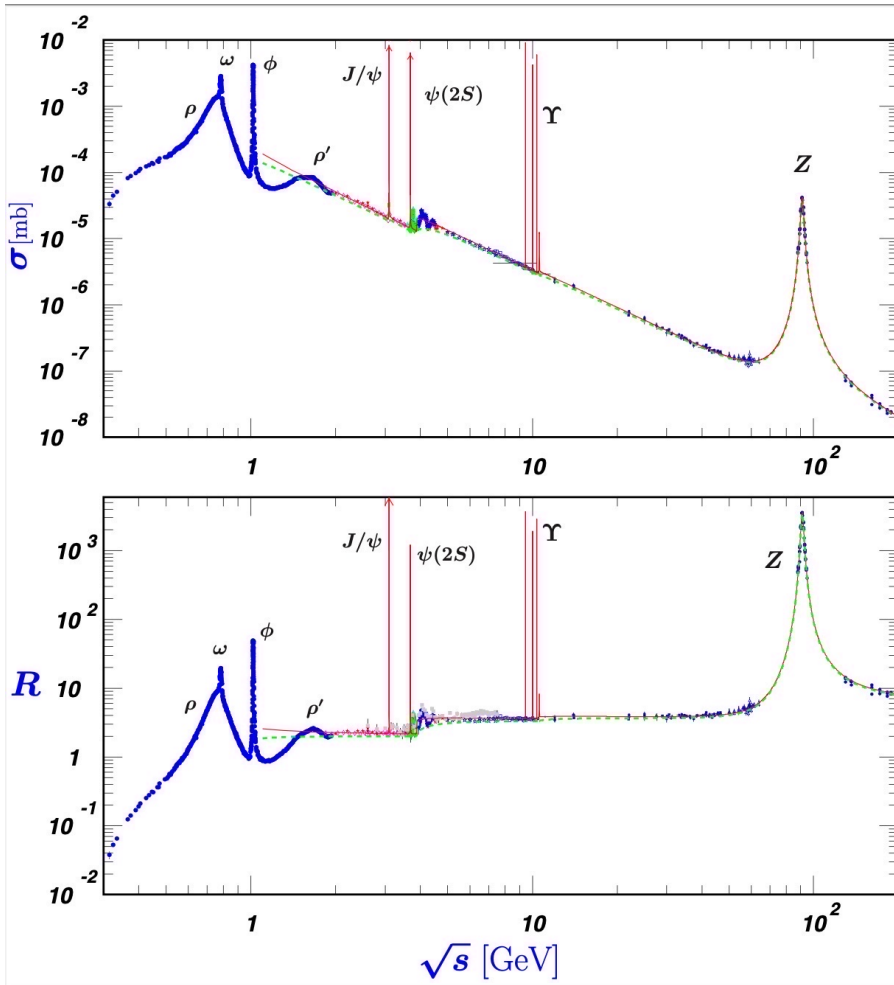
$$= \frac{\alpha^2}{4s} \left(\frac{3 + \cos^2 \theta}{1 - \cos \theta} \right)^2$$



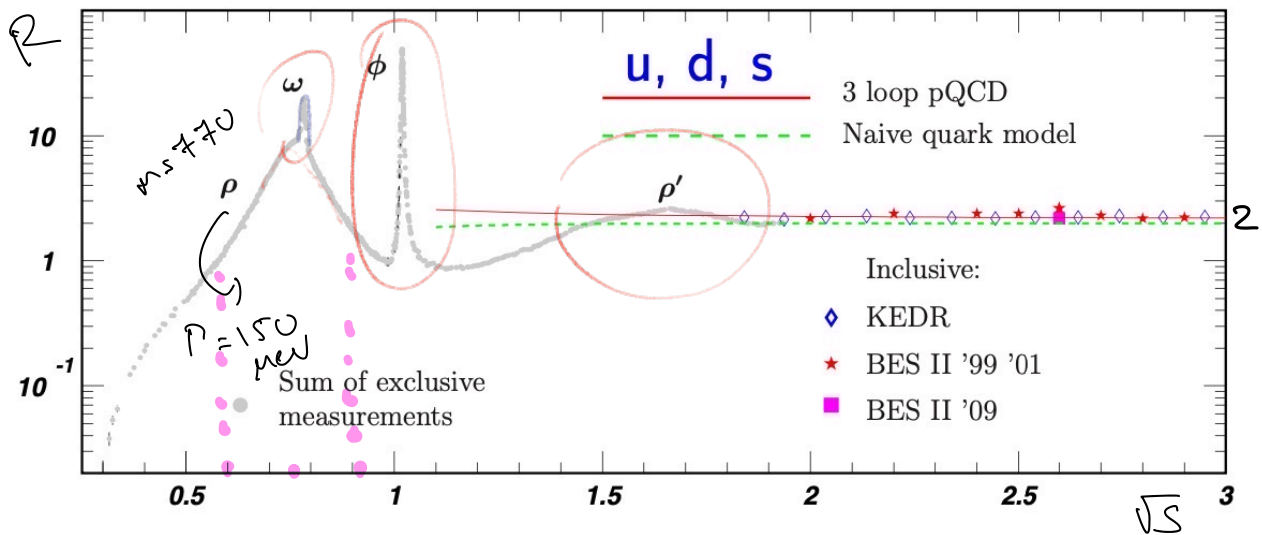
$$\frac{86.8 \text{ nb}}{s(\text{GeV}^2)}$$



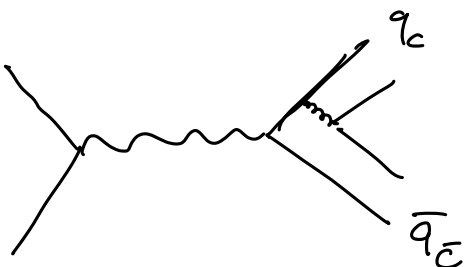
Ratio between measured cross section and QED calculation



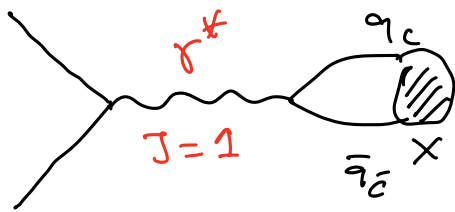
$$\sigma \propto \frac{1}{s}$$



why Resonances?



$$e^+e^- \rightarrow \text{hadrons}$$



m_X mass, Γ : total decay width.

$$J=1 \quad L=0 \Rightarrow S=1$$

π^0 not allowed

only ρ, ω

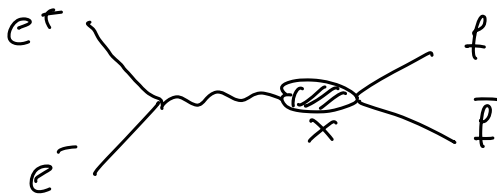
$$\omega: \quad m = 782 \text{ MeV} \quad \Gamma = 8 \text{ MeV}$$

$$\rho: \quad 770 \text{ MeV} \quad \Gamma = 150 \text{ MeV}$$

$$\phi(1020) \quad 1020 \text{ MeV} \quad \Gamma =$$

Cross section in presence of a Resonance

$$\sigma(e^+e^- \rightarrow X \rightarrow f\bar{f}) = \frac{16\pi}{s} \frac{2J+1}{(2S_a+1)(2S_b+1)} \frac{\Gamma_{ee}}{\Gamma_{\text{tot}}} \frac{\Gamma_{f\bar{f}}}{\Gamma_{\text{tot}}} \frac{\Gamma_{\text{tot}}^2}{(m_X - \sqrt{s})^2 + \Gamma_{\text{tot}}^2/4}$$



$$\Gamma_{ee} = \Gamma(X \rightarrow ee)$$

$$\Gamma_{f\bar{f}} = \Gamma(X \rightarrow f\bar{f})$$

J : tot. ang. mom of X

Γ_{tot} = total decay width

S_a : spin of f

S_b : spin of $\bar{f}$

$$\Gamma_{\text{tot}} = \Gamma_{ee} + \Gamma_{\mu\mu} + \Gamma_{f\bar{f}} + \dots$$

⑨ $\sqrt{s} < 3 \text{ GeV}$ R is ruined by wide resonances
of ρ, ω, ρ'

$$R=2 \quad \text{for } \sqrt{s} > m_{\rho'} + \Gamma_{\rho'}$$

$\rho(770)$

$$I^G(J^{PC}) = 1^+(1^{--})$$

$\rho(770)$ DECAY MODES

	Mode	Fraction (Γ_i/Γ)	Scale factor/ Confidence level
Γ_1	$\pi^+ \pi^-$	~ 100	%
Γ_2	$K^+ K^-$		

$\omega(782)$

$$I^G(J^{PC}) = 0^-(1^{--})$$

$\omega(782)$ DECAY MODES

	Mode	Fraction (Γ_i/Γ)	Scale factor/ Confidence level
Γ_1	$\pi^+ \pi^- \pi^0$	$(89.2 \pm 0.7) \%$	
Γ_2	$\pi^0 \gamma$	$(8.35 \pm 0.27) \%$	S=2.2
Γ_3	$\pi^+ \pi^-$	$(1.53^{+0.11}_{-0.13}) \%$	S=1.2

$\phi(1020)$

$$I^G(J^{PC}) = 0^-(1^{--})$$

$\phi(1020)$ DECAY MODES

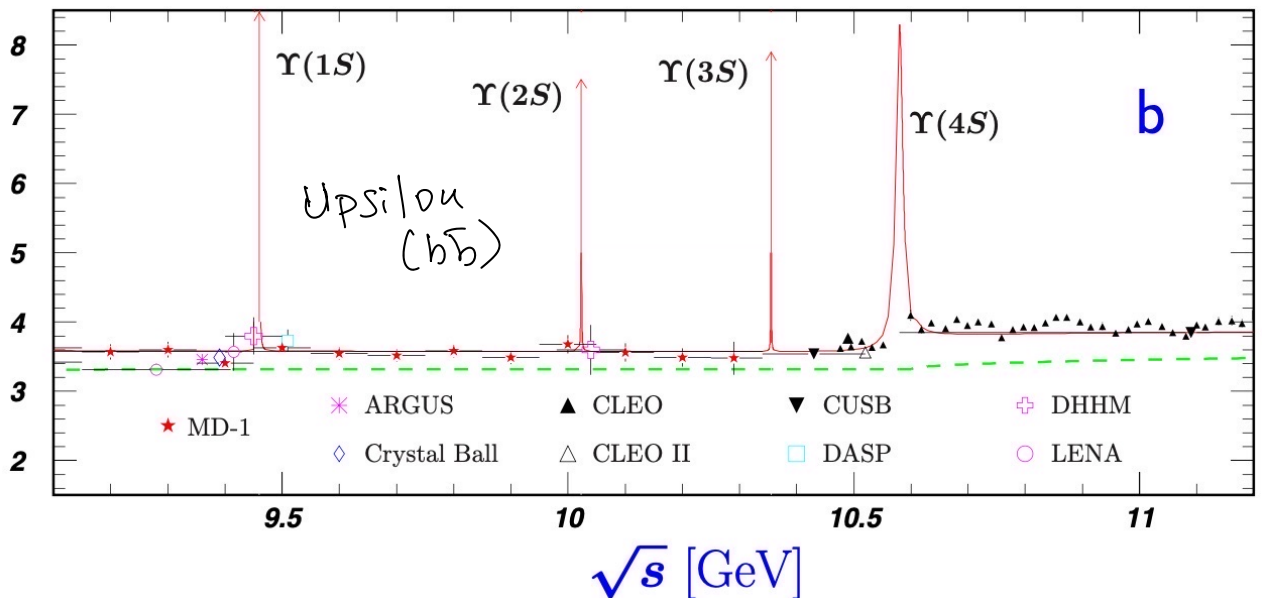
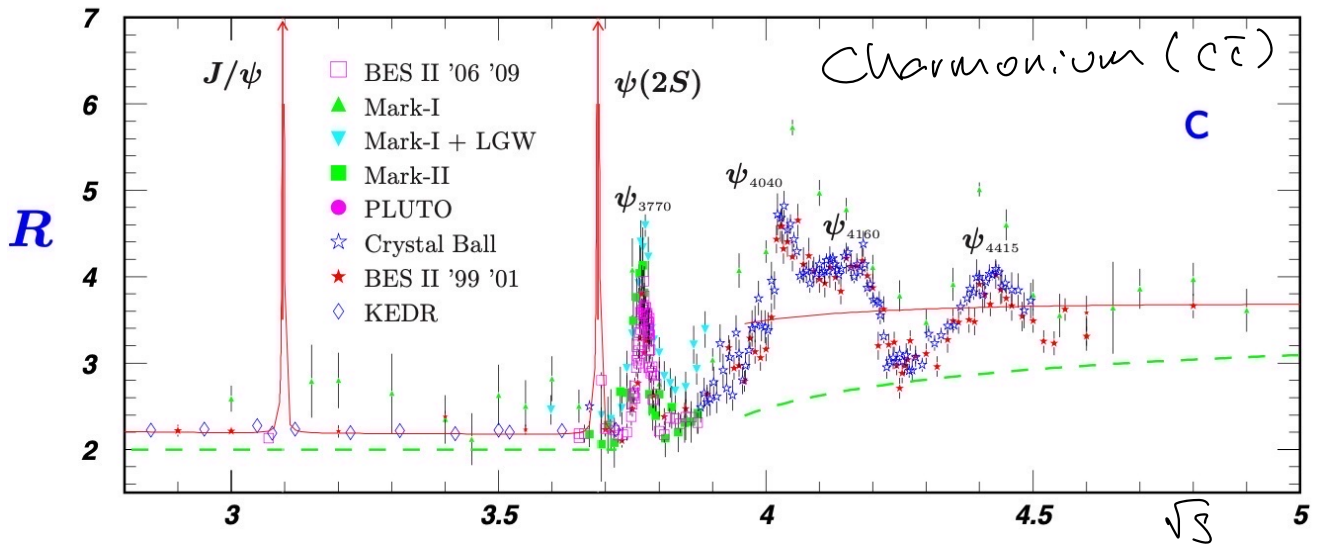
	Mode	Fraction (Γ_i/Γ)	Scale factor/ Confidence level
Γ_1	$K^+ K^-$	$(49.1 \pm 0.5) \%$	S=1.3
Γ_2	$K_L^0 K_S^0$	$(33.9 \pm 0.4) \%$	S=1.2
Γ_3	$\rho \pi^+ + \pi^+ \pi^- \pi^0$	$(15.4 \pm 0.4) \%$	S=1.2
Γ_4	$\rho \pi$		
Γ_5	$\pi^+ \pi^- \pi^0$		
Γ_6	$\eta \gamma$	$(1.301 \pm 0.025) \%$	S=1.2
Γ_7	$\pi^0 \gamma$	$(1.32 \pm 0.05) \times 10^{-3}$	
Γ_8	$\ell^+ \ell^-$	—	
Γ_9	$e^+ e^-$	$(2.979 \pm 0.033) \times 10^{-4}$	S=1.3
Γ_{10}	$\mu^+ \mu^-$	$(2.85 \pm 0.19) \times 10^{-4}$	

$\rho(1450)$

$$I^G(J^{PC}) = 1^+(1^{--})$$

$\rho(1450)$ DECAY MODES

	Mode	Fraction (Γ_i/Γ)
Γ_1	$\pi^+ \pi^-$	seen
Γ_2	$\pi^+ \pi^-$	seen
Γ_3	4π	seen
Γ_4	$\omega \pi$	
Γ_5	$a_1(1260) \pi$	
Γ_6	$h_1(1170) \pi$	
Γ_7	$\pi(1300) \pi$	
Γ_8	$\rho \rho$	
Γ_9	$\rho(\pi\pi)_{S\text{-wave}}$	



13.1 fb⁻¹ (13 TeV, 2016)

